

TrES-5b

R-0892

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250816 · created 2026-07-29T02:27:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460903.8435 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.054
Transit depth (Rp/Rs)^2	3.0 +/- 1.86 [%]
Orbital Inclination (inc)	80.37 +/- 2.83
Airmass coefficient 1 (a1)	0.17 +/- 0.16
Airmass coefficient 2 (a2)	1.31 +/- 0.99
Scatter in the residuals of the lightcurve fit is	89.35 %
Best Comparison Star	#1 - [324, 409]
Optimal Aperture	3.8
Optimal Annulus	10.77
Transit Duration (day)	0.039 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.054 vs 0.142 (deviation 0.32σ)
Duration fit vs published	0.039 d vs 0.0758 d (deviation 0.69σ)
Mid-time O–C	-8.9 min
Depth SNR	1.8
Residual scatter	89.35 %

Light curve

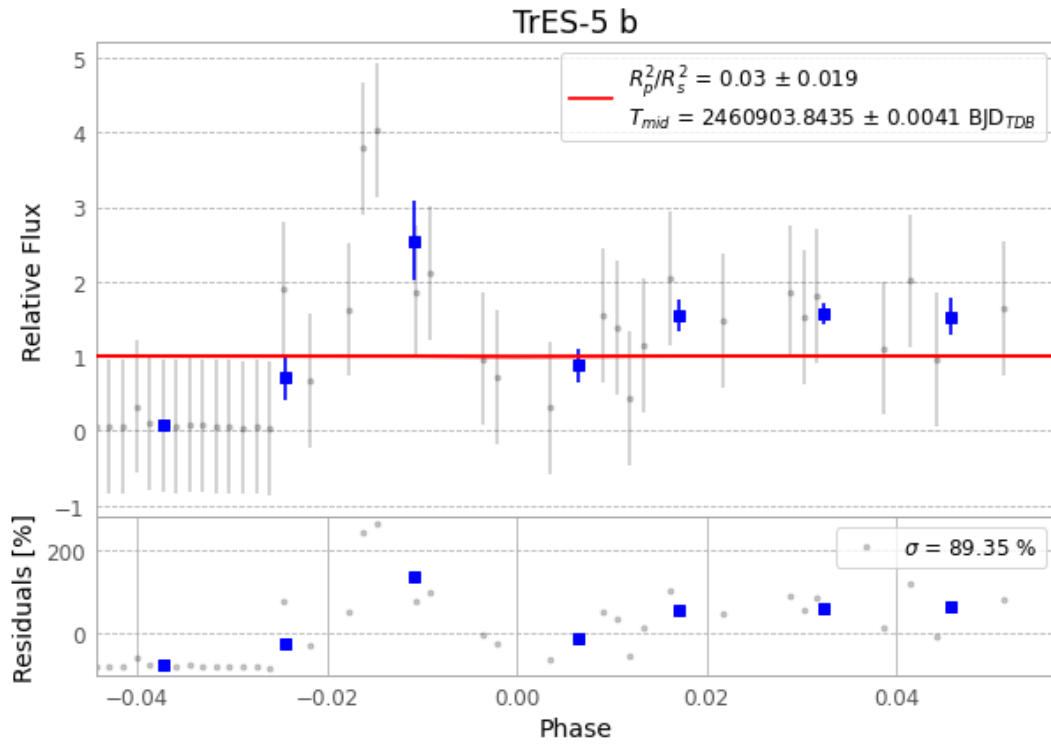


Figure 1 — FinalLightCurve_TrES-5 b_2025-08-15.png (SHA-256 b75a56fe48c9d084...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3f468fde9bd288ba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), TRES-5250816063615.FITS → TRES-5250816101215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250816.log (d367e404040b...), FinalLightCurve_TrES-5 b_2025-08-15.png (b75a56fe48c9...), FinalLightCurve_TrES-5 b_2025-08-15.csv (29c41a9bfd41...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 02:27Z.